

Kunj Rathod

Machine Learning Engineer – Production ML, Retrieval Systems, Model Evaluation

kunjrathod.com +1 (385) 202-8879 kunj.rathod@utah.edu linkedin.com/in/rathodkunj github.com/rathodkunj2005

Education

University of Utah

Aug 2023 – Dec 2026

B.S. in Computer Science

GPA: 3.7/4.0, Dean's List

Coursework: Machine Learning, Distributed Systems, Operating Systems, Computer Vision, NLP, Algorithms, Databases

Industry Experience

Software Engineering Intern – Microsoft Fabric (Azure Data)

May 2026 – Aug 2026

Microsoft, Redmond, WA

- Building the SKU Change History feature end-to-end (query builder, service layer, REST API) for the Fabric Manage Capacities admin portal, integrating internal telemetry pipelines to give capacity admins auditable SKU-change timelines across large enterprise tenants.
- Migrating capacity overage workflows into native Fabric platform UX with action-driven side panes; defining data contracts with UX and backend teams across 3 time zones.

Software Development Intern, AI Services (SUDO Program)

Jan 2025 – Present

University of Utah Health, Salt Lake City, UT

- Architected HIPAA-compliant LLM serving infrastructure for 90+ hospital executives: token-streaming inference at p95 <200ms TTFT and 40% latency reduction via pipeline optimization, API caching, and AWS Bedrock-backed model orchestration.
- Owned production APIs, schemas, dashboards-adjacent usage telemetry, and AWS infrastructure (Lambda, API Gateway, RDS, DynamoDB, S3); shipped 6 features across 4 sprints and implemented distributed session persistence for 1,000+ concurrent conversations.

AI Engineering Intern

Nov 2024 – Apr 2025

CourtEasy.ai / Nugen (Remote)

- Scaled hybrid retrieval (dense vectors + BM25 + reranking) over 10M+ documents, improving recall 28% and reducing hallucinations 35%; evaluated model quality with F1, latency, cost, and hallucination-rate metrics.
- Benchmarked 8 model families across LegalBench, NyayaAnumana, IL-TUR, and LegalBench-RAG; built model-routing analysis that reduced projected inference spend by \$50k+/yr while preserving answer quality.

Selected ML Systems & Research

BioGraphRAG – Distributed Biomedical Knowledge Graph System

May 2024 – Jan 2025

Python, FastAPI, NebulaGraph, LlamaIndex, Docker, AWS

- Engineered a distributed GraphRAG system over 1M+ biomedical entities (UniProt, AlphaFold, RXNav) with a 40% accuracy gain over embedding-only baselines; optimized graph traversal 3x via caching and node pruning (p95 <500ms) and built ETL pipelines processing 2M+ entity updates monthly.

MEMROT v2 – Mechanistic Evaluation of Long-Context Memory Failure

May 2026 – Present

Independent; PyTorch, Gemma-2-2B, Gemma Scope SAEs, LongMemEval-S, CHPC/SLURM

- Built a reproducible model-evaluation pipeline with gold-position randomization, counterfactual retrieval variants, SAE layer scans, mean-ablation, and intervention tests; ran an 840-pass sweep over 140 controlled retrieval questions on distributed CHPC/SLURM compute.

Circuit-Level Generalization via Sparse Feature Circuits

Jan 2026 – Present

University of Utah; PyTorch, attribution graphs, graph skeletonization

- Studying whether sparse feature circuits learned on default tasks generalize to counterfactual task variants; open-sourced a graph-skeletonization tool that prunes attribution graphs and exports visualization-compatible artifacts without dropping source metadata.

Continuum – Local-First Multimodal Memory Engine

Jun 2026

Rust, Tauri 2, React, TypeScript, LanceDB, OCR, embeddings, MCP

- Built a privacy-first desktop memory layer capturing foreground context on-device and storing vector + lexical retrieval fields in LanceDB, exposing agent-ready recall through an MCP server.

Languages & Technologies

Languages: Python, Java, C++, Rust, TypeScript, SQL

ML: PyTorch, TensorFlow, transformers, model evaluation, offline metrics, retrieval/ranking, RAG/GraphRAG, vector search, BM25, reranking, embeddings, sparse autoencoders

Production & Data: AWS, Azure, Docker, Kubernetes, FastAPI, REST APIs, distributed systems, telemetry pipelines, SLURM, NebulaGraph, LanceDB, PostgreSQL